

SOURAV LAHON

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EDUCATION

National Institute of Technology Silchar
Bachelor of Technology in Computer Science and Engineering

Cachar, Assam, India
2021 – 2025

EXPERIENCE

Software Engineer

BlackRock

Jul 2025 – Present

Gurgaon, Haryana, India

- Develop and maintain data transformation tools in Perl, Python, and SQL to convert raw financial data into client-ready deliverables; conduct risk-return analysis to ensure market data accuracy and consistency.
- Automate the index creation process via scripting, cutting build time from **30 minutes to 5 minutes** across fixed income, equity, composite, and hedge index types.
- Build a real-time data ingestion pipeline in C#, .NET, and Kafka, applying NLP to extract and structure financial information from unstructured data sources.

Data Analyst Intern

BlackRock

Jan 2025 – Jun 2025

Gurgaon, Haryana, India

- Built an end-to-end automation application in Python, Pandas, Unix, and Streamlit that generated and deployed configuration files to remote hosts via a custom algorithm, reducing deployment time from 4–5 minutes to under 30 seconds (**90% reduction**).
- Developed Python and SQL scripts to automate quality control reporting across market indexes, incorporating custom algorithms for data validation and structured report generation, reducing **80%** of manual effort.

Software Engineer Intern

Microsoft

May 2024 – Jul 2024

Hyderabad, Telangana, India

- Created a debug tool for the MSAI-UX team, resulting in a **30%** reduction in triage time.
- Engineered features including Response Debugger, Accessibility Debugger, and Performance Analyzer, contributing to a **50%** performance increase.

PROJECTS

Assam Dress | *MongoDB, Express, React, Node* | [website](#)

- Built an e-commerce site to promote Assamese culture, reaching **100+** visitors in the first month.
- Implemented secure user authentication with OTP, password hashing, and JWT recovery, reducing login errors by **40%** during testing.
- Implemented CRUD operations with Axios and Redux, creating a fluid user interface that streamlined data management for **20+** users during the testing phase.
- Integrated Cloudinary for storing **200+** product images and Stripe for payment processing.
- Integrated Redis for session management, caching frequently accessed product data, cart storage, and rate limiting, improving API response times and system reliability under concurrent user load.

Audio-Visual Speech Recognition (AVSR) | *Machine Learning, Python* | [thesis](#)

- Designed a multi-modal speech recognition system improving recognition accuracy by **25%** in noisy environments compared to traditional audio-only systems.
- Fused audio and video streams using MFCCs and CNNs, resulting in a **20%** boost in speech recognition accuracy under noisy conditions.
- Engineered and evaluated classifiers including SVM, Random Forest, DNN, and LSTM, achieving a **30%** improvement in recognition rates in challenging acoustic settings.
- Refined feature extraction and data fusion techniques, improving model robustness by **15%** and enabling real-time interaction with under **500ms** latency in noisy environments.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, JavaScript, Perl, SQL, HTML, CSS

Frameworks & Libraries: React, Redux, React Native, .NET, Node.js, Express.js, NestJS, Flutter, Pandas, Streamlit

Developer Tools: Git, Unix, Kafka, Docker, Redis, MongoDB, Stripe, Cloudinary, Power BI

Concepts: Data Structures, Algorithms, OOP, REST APIs, NLP, Machine Learning, Deep Learning, Event-Driven Architecture, JWT Authentication, Agile/Scrum

CERTIFICATES

Microsoft Learning Path | [Data Analysis](#), [Generative AI](#), [GitHub Professional](#)

- Earned **3** career essentials certifications covering data analysis methods, generative AI concepts, and professional GitHub workflows by Microsoft and GitHub.

Software Engineer Internship | [Certificate by Microsoft](#)

- Awarded upon successful completion of the MSAI-UX team internship program, recognized for production debug tool delivery.